Contents

1 Appendix D: Neural Encoding Efficiency on Time-Series

1.1 Objective

Comprehensive neural encoding analysis including spike-train generation, temporal dynamics, population coding, mutual information, and adaptation mechanisms for olfactory receptor neurons.

1.2 Methods (src)

- `src/case_studies/neural_encoding.py`
  - `generate_spike_trains(stimuli, dt, baseline_rate, max_rate, dynamics)` — realistic spike generation
  - `analyze_spike_train_statistics(spike_data)` — ISI, CV, Fano factor
  - `temporal_coding_analysis(spike_data, stimulus_times)` — latency and precision metrics
  - `population_coding_analysis(population_responses, labels)` — PCA, LDA, correlation structure
  - `mutual_information_analysis(responses, stimuli)` — information-theoretic metrics
  - `odor_discrimination_analysis(responses, odor_ids, time_windows)` — discrimination performance
  - `adaptation_dynamics_analysis(spike_data, stimulus_duration)` — adaptation characterization
  - `information_rate_time_series(responses, dt_s, noise_std)` — channel-capacity estimation
  - `rate_coding_metrics(responses, labels)` — separability and discriminability metrics

1.3 Script and outputs

- Script: `scripts/generate_neural_encoding_analysis.py`
- Data: `output/data/neural_encoding_comprehensive.npz`
- Figure: `figures/neural_encoding_comprehensive_analysis.png`

1.4 Figure

1.5 Equation references

- Information rate: see (**??**)
- Response time model: see (**??**)

[width=0.9]figures/neural$_{e}ncoding_{c}omprehensive_{a}nalysis.png$

Figure 1: *Neural encoding analyses generated deterministically by* $‘scripts/generate_{n}eural_{e}ncoding_{a}nalysis.py‘using‘src/case_{s}tudies/neural_{e}ncoding.py‘.Panelsincludespiketrains, tempo$

[width=0.9]figures/integrated$_{a}nalysis_{i}nformation_{a}nalysis.png$

Figure 2: *Integrated information analysis (from* $‘scripts/generate_{i}ntegrated_{a}nalysis.py‘)showingmolecular, receptor, neuralandenvironmentalinformationdecompositio$ $domaininformationbalances.$

1.6 Reproducibility

- Run: `python3 scripts/generate_neural_encoding_analysis.py`
- Artifacts saved to `output/data/` and `figures/`.
- Deterministic seeds: `src/config.set_random_seed(42)` for surrogate time-series.

1.7 Cross-references

- Methods: sec:methodology
- Symbols: sec:symbols$_{g}lossaryMathappendix : sec : mathematical_{a}ppendix$